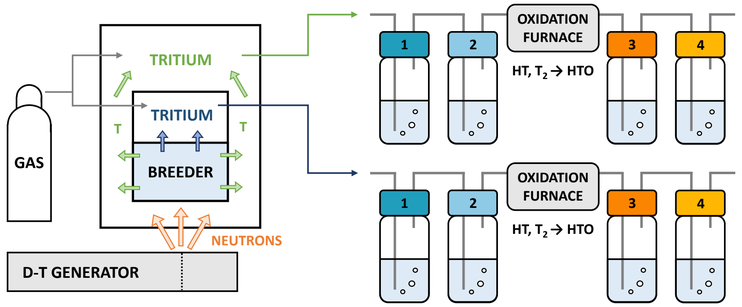
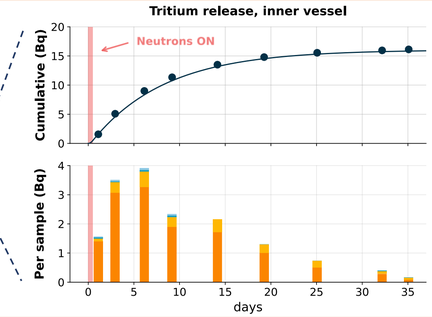
Breeding tritium in your lab?

The BABY recipe:

1. Add **0.1-1 L of breeder** to a crucible and place the crucible in a secondary container
2. Heat to **500-700°C**
3. Flow dry, oxygen-free **gas** through the inner and outer vessel
4. Irradiate with **14 MeV neutrons** at 10^8-10^9 n/s for 4-12 h





5. Place two sets of water **bubblers** downstream, each containing 2x2 vials separated by a furnace for **catalytic oxidation**
6. Let the gas bubble during and after irradiation; **sample occasionally**
7. Count collected samples using **Liquid Scintillation Counting**

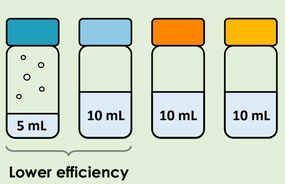
Here is how to measure it:

1 Select adequate sample volume and bubbling time, based on tritium production and scintillation cocktail

Small-scale breeding experiments produce **10s-100s of DPM of tritium** per irradiation. For accurate accountancy all collected tritium should also be counted.

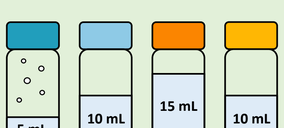
A 20 mL LSC sample can tolerate up to **10 mL of the aqueous component** with ~25% counting efficiency[1].





Dry gas leads to **evaporation**[2] from the first vial, reducing collection efficiency. Limit bubbling time to ~5 days, or periodically replenish water in the first vial.

When the gas contains **hydrogen**, water level in the 3rd vial will rise as H_2 is oxidized in the furnace. Divide the sample before counting, or start with a lower collection volume.

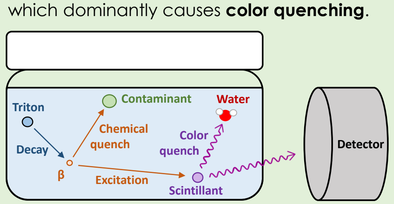


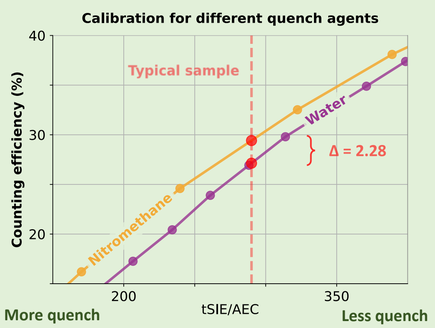
[1]When using UltimaGold LLT scintillation cocktail.
[2]Cca. 1 mL/day at 30 SCCM of dry He.

2 Create a custom quench calibration for your specific sample preparation

LSC counters come with pre-installed quench curves usually prepared with standards quenched with **nitromethane**. These are **chemically quenched** standards.

The primary quenching agent in samples prepared in breeding experiments is **water**, which dominantly causes **color quenching**.





A typical sample in these experiments has a 1:1 water-to-cocktail ratio. If the built-in calibration is used, the amount of tritium released over a typical run would be **undercounted by ~8%**.

3 Avoid or mitigate sample contamination

Bubblers may collect **contaminants from the gas**: particulates, breeder decomposition products, off-gas from heating elements. Contamination increases quenching of prepared LSC samples.

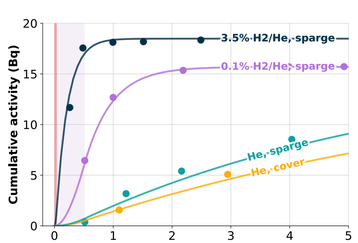


Avoid quenching by filtering sample gas or **neutralizing acidic samples**. If a known and measurable contaminant can not be eliminated, create set of quenched standards to **re-calibrate the counter** for the modified sample chemistry.

A robust tritium measurement procedure yields:

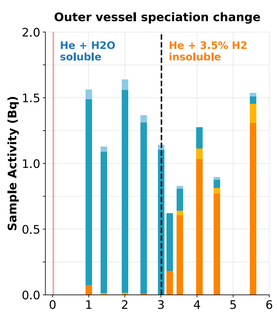
a Minimum detectable activity of ~0.07 Bq per sample

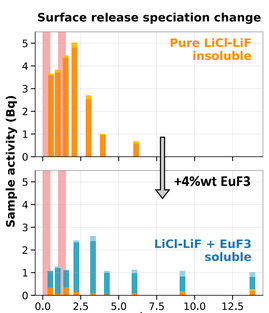
With an adequate sampling schedule, a low MDA enables accurate tritium accountancy and gives detailed insight into **tritium release dynamics**.



b Insight into the chemical speciation of tritium

The bubbler furnace enables separating tritium species, allowing the study of factors influencing **radiochemistry**, such as impurities in the breeder or the composition of the gas.

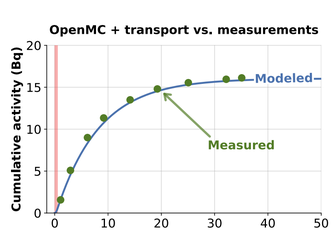




c Accurate total production measurements

Full tritium accountancy enables comparison between the measured **Tritium Breeding Ratio (TBR)** and values obtained from neutronics models.

$$\mathrm{TBR} = \frac{\text{tritons created}}{\text{tritons consumed}}$$



We are committed to open science principles and reproducibility.

Raw data and analysis available at: ***github.com/LIBRA-project***

LIBRA

Presented by Nikola Goleš (goles@mit.edu) on behalf of the LIBRA group at MIT PSFC.

Recent publication on BABY: *bit.ly/100mL-BABY*

LIBRA project overview: *bit.ly/LIBRA-overview*

DOE ARPA-E awardee (DE-AR0001542). The views and opinions of authors expressed herein do not necessarily state or reflect those of the U.S. Government or any agency thereof.

ARPA-E

MIT Plasma Science & Fusion Center